EECS708: Machine Learning

Clustering

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Slide thanks: Dr. Tim Hospedales

Course Context

- Supervised Learning
  - (Linear) regression
  - Logistic Regression (Classification)
  - Neural Networks
- Unsupervised
  - Clustering
  - Density Estimation
  - Dimensionality reduction (partial)
- Advanced topics
  - Deep Learning, Convolutional Neural Networks
  - Ensemble Learning

Unsupervised Learning Motivation

- Unsupervised Learning:
  - Too much data: We need to save memory/computation.
    - Reduce the data to a more manageable amount
  - Don’t understand the data
    - Exploratory data analysis
    - What underlying knowledge is there?
    - Discover patterns & trends
- Clustering
  - Focus on instances (rows)
- Dimension reduction
  - Focus on dimensions (columns)

Overview

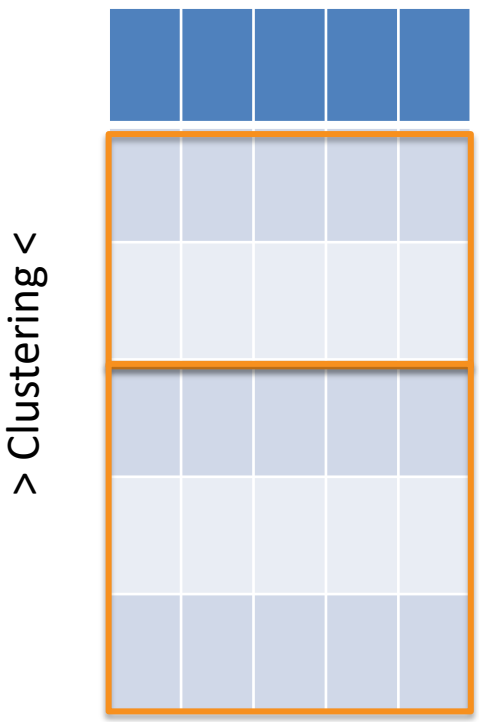
- Overview: What is clustering about?
- Clustering Algorithms
  - K-means
  - Hierarchical Clustering
- Further topics
  - Choosing the number of clusters
  - Advanced algorithms
- Some applications
  - ..besides marketing!

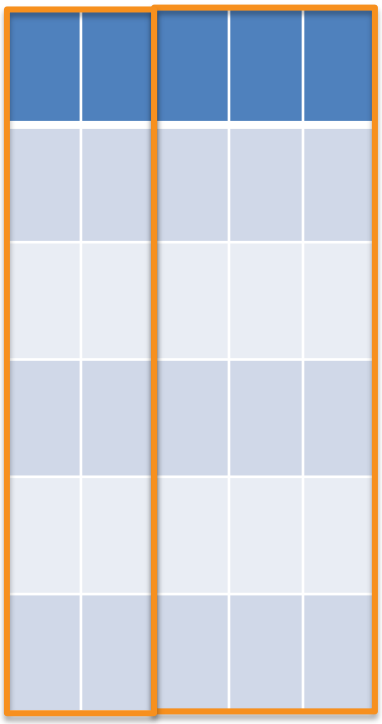
Clustering: Unsupervised Learning Context

Clustering versus Dimensionality Reduction

- Both unsupervised
- Group rows versus columns

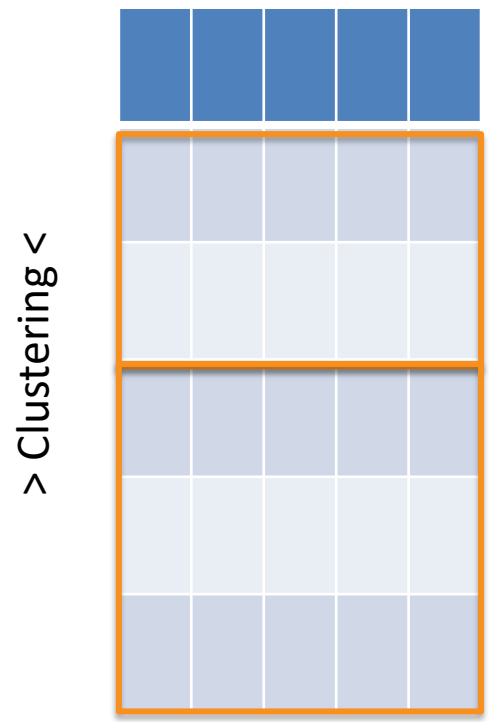
> Dimensionality Reduction <



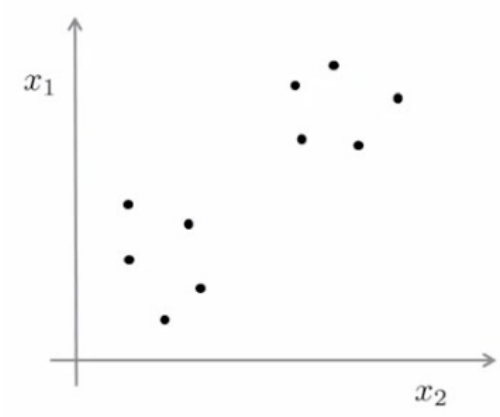


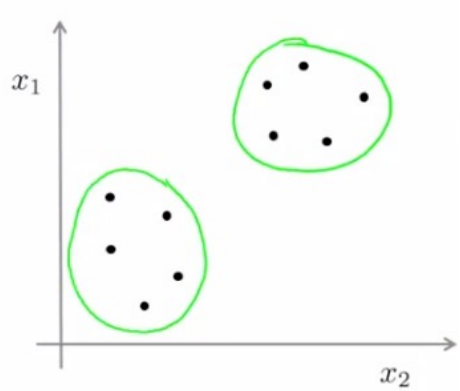
Clustering

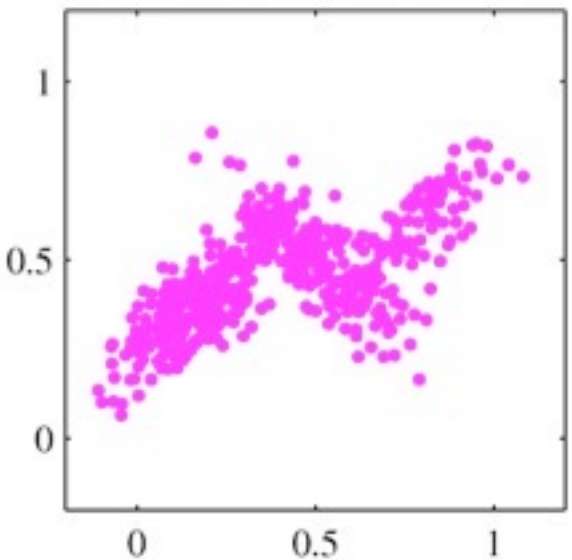
- Problem definition:
  - Given a set of examples $(x_1, x_2, .., x_N)$
  - Divide them into subsets of similar examples.
- Alternative view:
  - How to do classification if you have no labeled training examples
- Conceptual & algorithmic challenges:
  - How to quantify similarity?
  - How to measure the goodness of a partition?

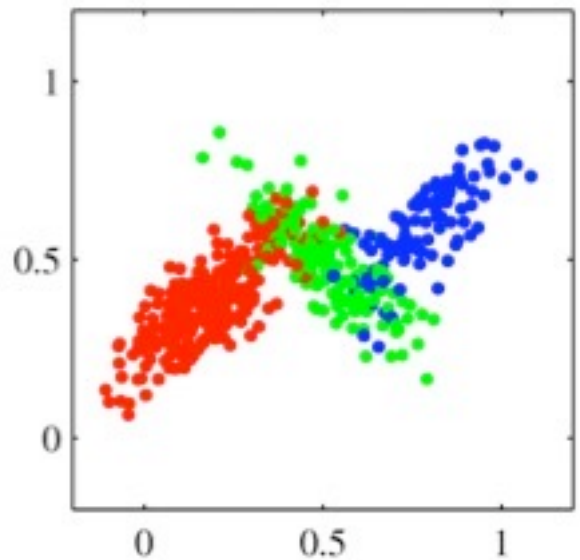


Clustering: Visual Example









Clustering: Why do it?

- Market segmentation
  - Teenagers, Mothers, Empty-nesters
  - Targeted products/marketing for each “cluster” of customers
- Data-center organization
- Social network analysis (find recommended friends)
- Group articles on your website or blog
- Group websites on your aggregator
- T-shirts: Given height & weight in the population:
    - How big should size of S, M, L be?
    - Should you offer S,M,L or XS, S, M, L, XL?

Clustering: Why do it?

- T-shirts: Given height & weight in the population:
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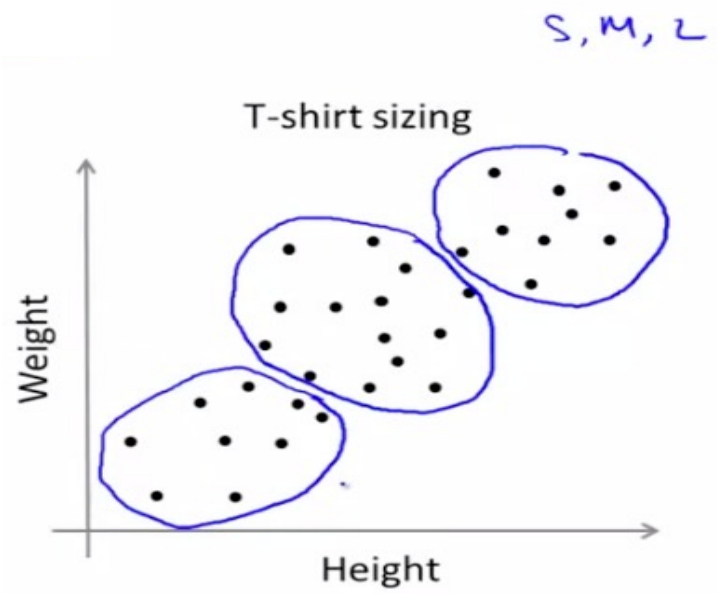
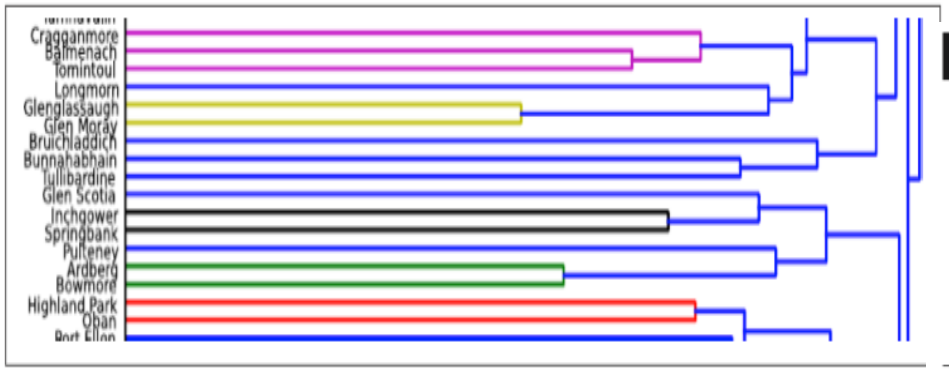


Clustering: Why do it?

- Stock choice in retail: There are more possible products to stock than we have space for. Which ones to include?
  - Try to cover the range of possible options so a consumer of every preference can find something they like.
  - How? Clustering then pick one example from each cluster.
  - Question: Have you covered the space well?



Whiskey	Distance	Descriptors
Bunnahabhain	—	*gold; firm,med,light; sweet,fruit,clean; fresh,sea; full*
Glenglassaugh	0.643	gold; firm,light,smooth; sweet,grass; fresh,grass
Tullibardine	0.647	gold; firm,med,smooth; sweet,fruit,full,grass,clean; sweet; big,arome,sweet
Ardberg	0.667	sherry; firm,med,full,light; sweet; dry,peat,sea;salt
Bruichladdich	0.667	pale; firm,light,smooth; dry,sweet,smoke,clean; light; full
Glenmorangie	0.667	p.gold; med,oily,light; sweet,grass,spice; sweet,spicy,grass,sea,fresh; full,long

Summary

- What is clustering about?
- Motivation for organization of the data
- Applications

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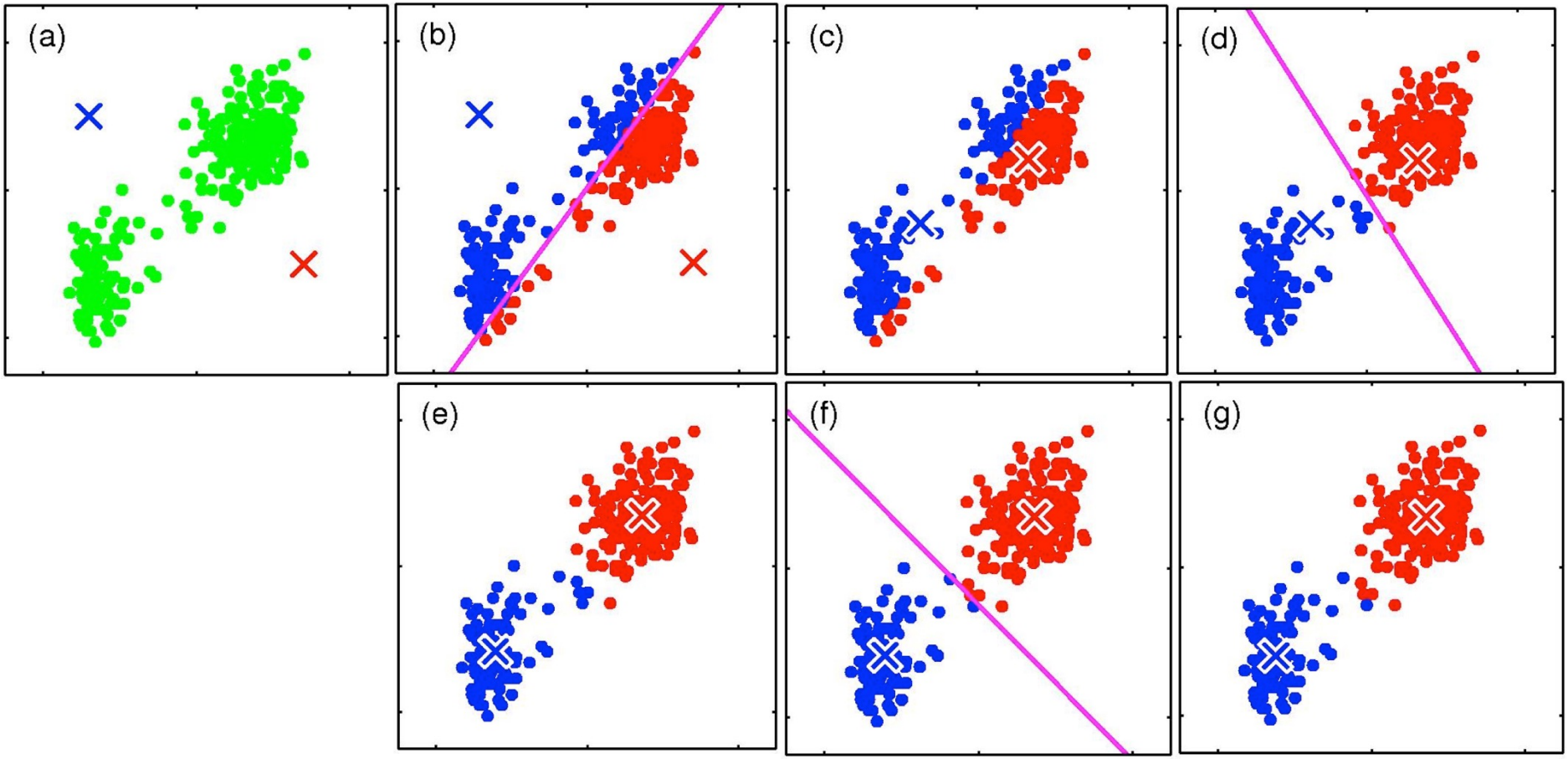
K-Means Algorithm - illustration



K-Means Algorithm

- Input:
  - N point dataset, $X=\{x_1,x_2,\ldots x_n\}$,
  - Number of clusters K.
- Initialize randomly K centers $\mu_1, \mu_2, \ldots, \mu_K$
- Repeat
  - For i=1:N
    - Labels_i=Cluster centroid closest to x_i
  - For k=1:K
    - μ_k = average of points assigned to cluster k

Formalizing K-means

- Cost Function:
  - Find cluster centers $\mu_{1:k}$ and cluster assignments $c_{1:N}$ so as to minimize the sum squared distances of points from assigned clusters:

$$E_{KM}(X, c_{1:N}, \mu_{1:K}) = \sum_{i=1}^{N} \left(x_i - \mu_{c_i}\right)^2$$

- Algorithm:
  - "E-step": Find cluster closest to each point (fix $\mu_{1:k}$ and optimise $c_{1:N}$)
  - "M-step": Find new center of each cluster (fix $c_{1:N}$, minimize for $\mu_{1:k}$)
- Aside:
  - We have seen algorithms with exact & gradient solutions to problems.
  - This is our first alternating minimization solution
  - An exact solution to each part of the problem given the other: Iterate

K-means - pseudocode

Find cluster centers $\mu_{1:k}$ and cluster assignments $c_{1:N}$

$$E_{KM}(X, c_{1:N}, \mu_{1:K}) = \sum_{i=1}^{N} \left(x_i - \mu_{c_i}\right)^2$$

Data structures:

D is #dinemsions, N is #samples, K is #clusters

X (data matrix)	NxD	reals
c (cluster assignments)	N	integers in {1..K}
μ (cluster centres)	KxD	reals

Algorithm:

- "E-step": Find cluster closest to each point (fix μ, minimize for c)
- "M-step": Find new center of each cluster (fix c, minimize for μ)

K-means - pseudocode

Find cluster centers $\mu_{1:k}$ and cluster assignments $c_{1:N}$

$$E_{KM}(X, c_{1:N}, \mu_{1:K}) = \sum_{i=1}^{N} \left(x_i - \mu_{c_i}\right)^2$$

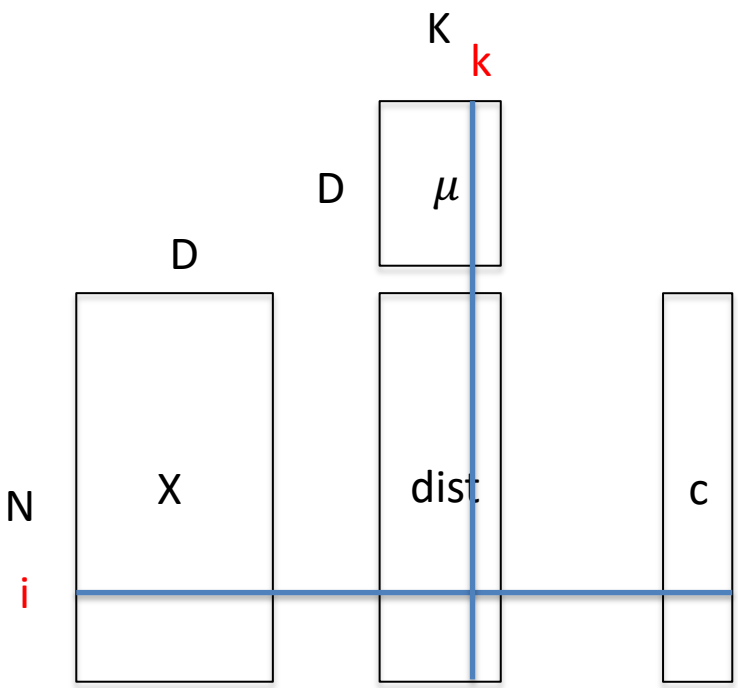
Data structures:

X (data matrix)	NxD	reals
c (cluster assignments vector)	N	integers in {1..K}
μ (cluster centres matrix)	KxD	reals

Algorithm:

– "E-step": Find cluster closest to each point (fix μ, minimize for c)

```
For 1< i < N
        for 1<k<K
                distance[I,k] = L2_norm(X[i,:]- μ[k,:]];
        end
        c[i] = argmin_k(distance[i,k]);  % c[i] is the index of the closest cluster centre
end
```



Algorithm:

- "E-step": Find cluster closest to each point (fix μ, minimize for c)

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K-means - pseudocode

Find cluster centers $\mu_{1:k}$ and cluster assignments $c_{1:N}$

$$E_{KM}(X, c_{1:N}, \mu_{1:K}) = \sum_{i=1}^{N} \left(x_i - \mu_{c_i}\right)^2$$

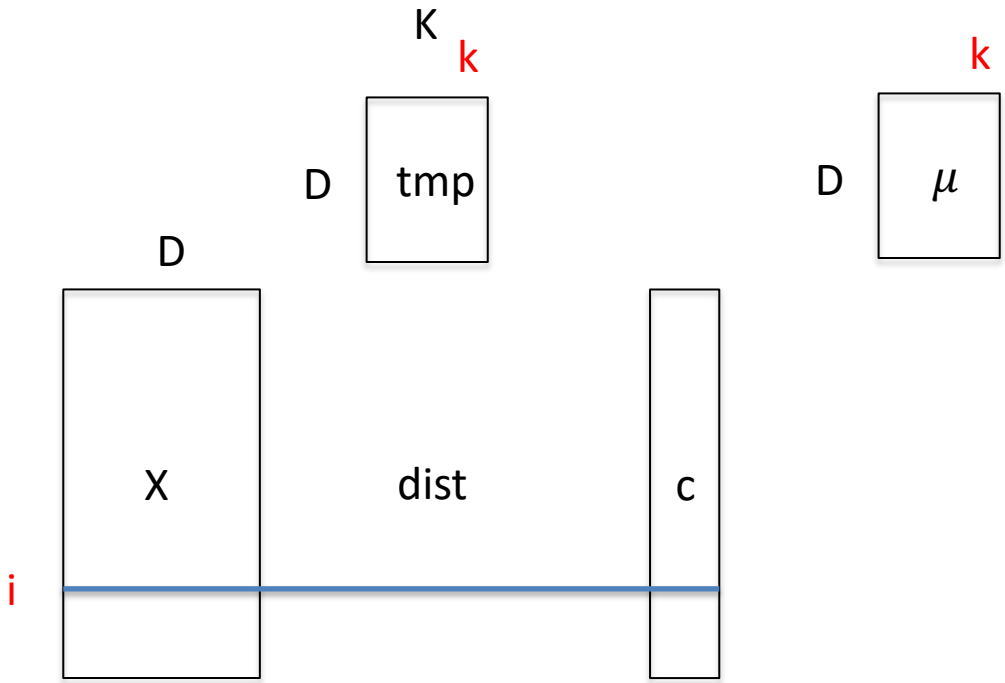
Data structures:

X (data matrix)	NxD	reals
c (cluster assignments vector)	N	integers in {1..K}
μ (cluster centres matrix)	KxD	reals

Algorithm:

– M-step: "Find new center of each cluster (fix c, minimize for μ)

```
tmp  = zeros(K,D);
tmpcount = zeros(K);              How many points in cluster K
For 1< i < N
      tmp[c[i],:] = tmp[c[i],:]  + X(i,:];  tmpcount[c[i]] += 1;
 End
 For 1 < k< K
       μ[k,:] = tmp[k,:] / tmpcount[k];
 end
```



Algorithm:

– M-step: "Find new center of each cluster (fix c, minimize for u)

```
tmp  = zeros(K,D);
tmpcount = zeros(K);                How many points in cluster K
For 1< i < N
        tmp[c[i],:] = tmp[c[i],:]  + X(i,:];  tmpcount[c[i]] += 1;
 End
 For 1 < k< K
         μ[k,:] = tmp[k,:] / tmpcount[k];
 end
```

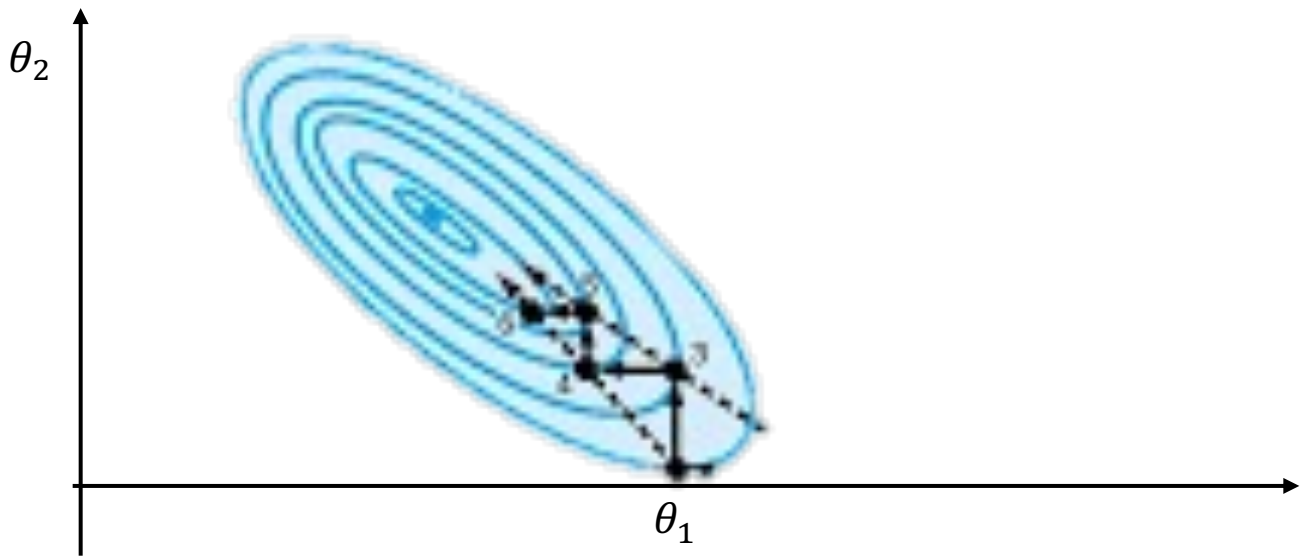
Coordinate descent

Minimize $J(\theta_1, \theta_2)$ by iterating between two steps:

Step 1: Keep θ_2 fixed and optimise w.r.t. θ_1 , i.e., $\theta_1^{k+1} = argmin\, J\left(\theta_1, \theta_2^k\right)$

Step 2: Keep θ_1 fixed and optimise w.r.t θ_2, i.e., $\theta_2^{k+1} = argmin\, J\left(\theta_1^{k+1}, \theta_2\right)$

Until convergence



Given some conditions, the procedure converges to a local minimum

“Optimality” of K - means (1)

Cost function $J(c_1, \ldots, c_N, \mu_1, \ldots, \mu_K) = \sum_n \|x_n - \mu_{c_n}\|^2$

Step 1: Optimise w.r.t $c_1, \ldots, c_N$

But each of the $c_1, \ldots, c_N$ appear only in one of the terms of the sum in the cost function, therefore we can minimize w.r.t. each individually

$$c_n = \mathrm{argmin}_{c_n} \sum_{n=1} \|x_i - \mu_{c_i}\|^2 = \mathrm{argmin}_k \|x_n - \mu_k\|^2$$

Thus, to minimise the cost with respect to classifications we choose $\mathbf{c_n}$ to be the cluster k for which $||x_n - \mu_k||^2$ is smallest (i.e. we choose the closest centre).

“Optimality” of K - means (1)

Cost function $J(c_1, \dots, c_N, \mu_1, \dots, \mu_k) = \sum_{n=1}^{N} \left|\left|x_n - \mu_{c_n}\right|\right|^2$

Step 2: Optimise w.r.t μ_k by setting the derivative wrt it to zero.

$$\nabla_{\mu_k} J(\dots) = \sum_{x_n \in C_k} -2\mu_k(x_n - \mu_k)$$

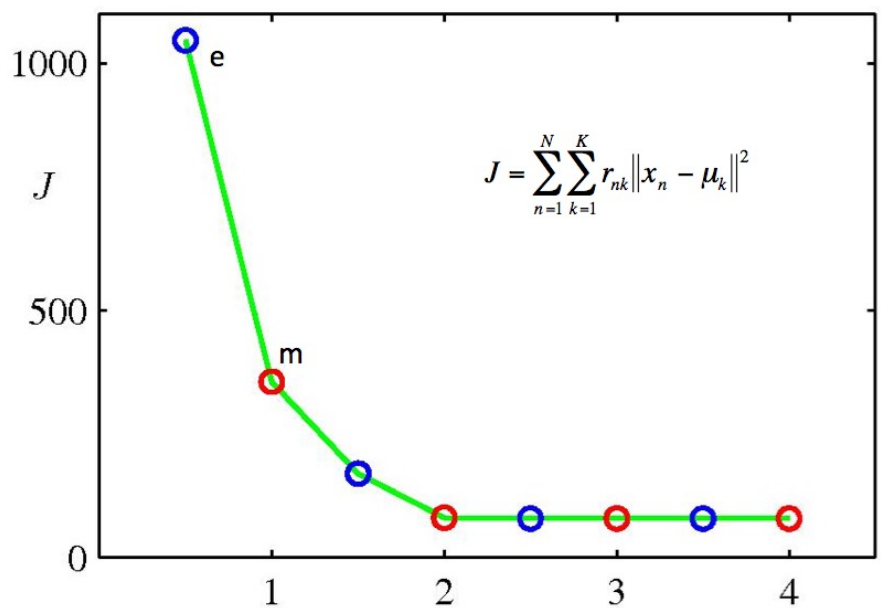
$$\nabla_{\mu_k} J(\dots) = 0 \Rightarrow \mu_k = \frac{\sum_{x_n \in C_k} x_n}{\sum_{x_n \in C_k} 1} = \frac{1}{|C_k|} \sum_{x_n \in C_k} x_n$$

Where $|C_k|$ are the number of points assigned to cluster k
That is, the μ_k that minimizes the cost function is the average of the points that belong to cluster k.

Formalizing K-means

$$E_{KM}(X, c_{1:N}, \mu_{1:K}) = \sum_{i=1}^{N} \left(x_i - \mu_{c_i}\right)^2$$

- Cost Function:
  - Find cluster centers $u_{1:k}$ and cluster assignments $c_{1:N}$ so as to minimize the distance of each point from it's assigned cluster:



K-means Properties

- Recall the algorithm:
  - Repeat S times:
  - "E-step": Find cluster closest to each point (fix μ, minimize for c)
  - "M-step": Find new center of each cluster (fix c, minimize for μ)
- Computation time?
  - O(NK), Or O(NKDS) if dimension and iterations included
  - Fast relative to O(N^2), slow relative to O(N). (i.e., if large K)

K-means Properties

- Distance Metric
  - Typically use Euclidean
    - May or may not be appropriate depending on data.
    - May not be robust to outliers
  - What happens if you have categorical data?
    - use 1-of-N encoding

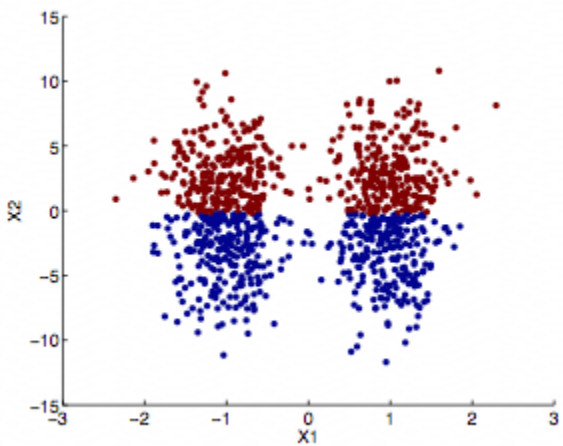
$$E_{KM}(X, c_{1:N}, \mu_{1:K}) = \sum_{i=1}^{N} \left(x_i - \mu_{c_i}\right)^2$$

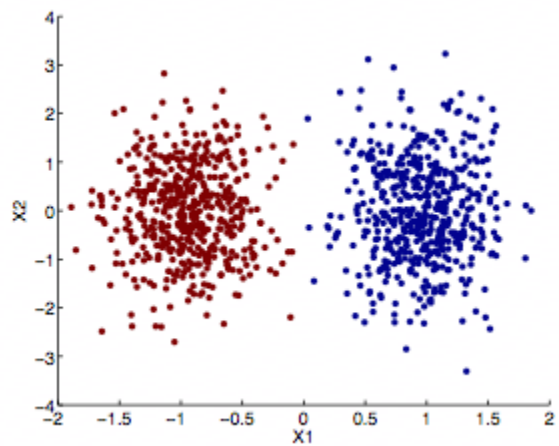
- Convergence:
  - It converges to a local minima only
  - => In practice repeat with many random initializations and pick the best
  - Different distances lead to changes in both steps!!

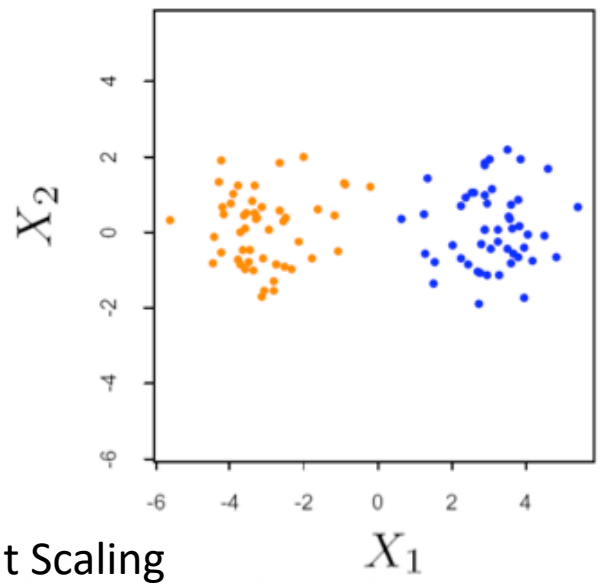
K-means: When it (doesn’t) work

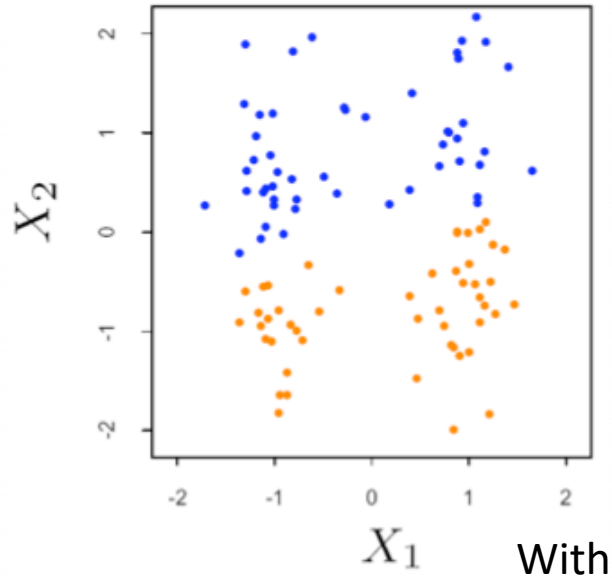
- Sensitive to data scaling
  - Renormalize in [0,1] or by standard deviation

Scaling?







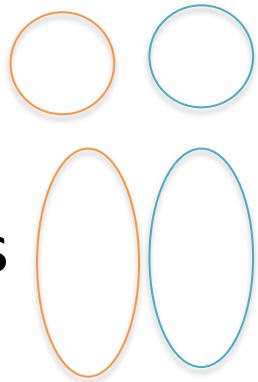


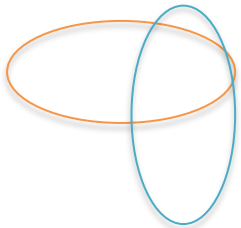
Without Scaling

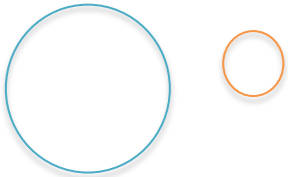
With Scaling

K-means: When it (doesn't) work

- Works if:
  - Clusters are spherical
  - Clusters are well separated
  - Clusters are of similar volumes
  - Clusters have same number of points







Summary

- K-means groups data into K groups
- K-means as an iterative algorithm that that optimizes a cost
- K-groups of data – K given.
- Influence of scaling
- When k-means work

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- Some applications
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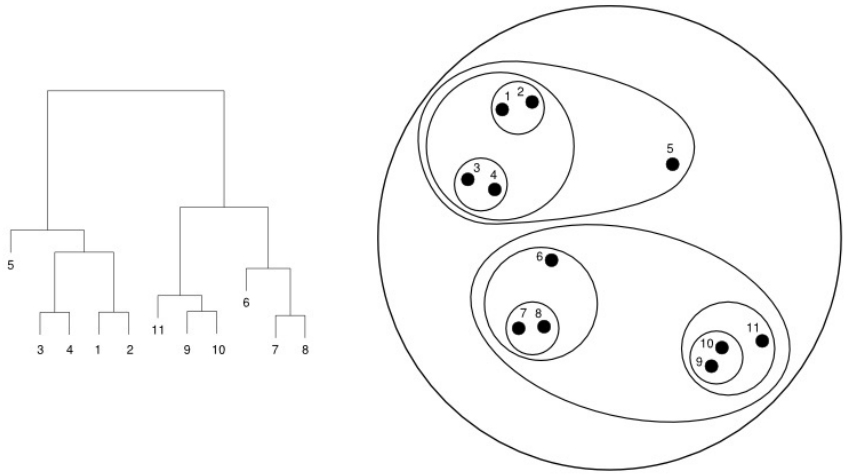
Hierarchical Clustering

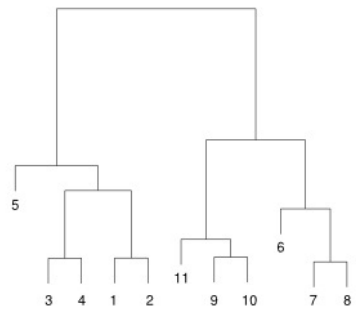
- Sometimes you want a tree of similarity rather than a flat clustering
  - (And K-means clusters discovered can be sensitive to chosen K)
- Output: A dendrogram (instead of cluster centers and assignments)

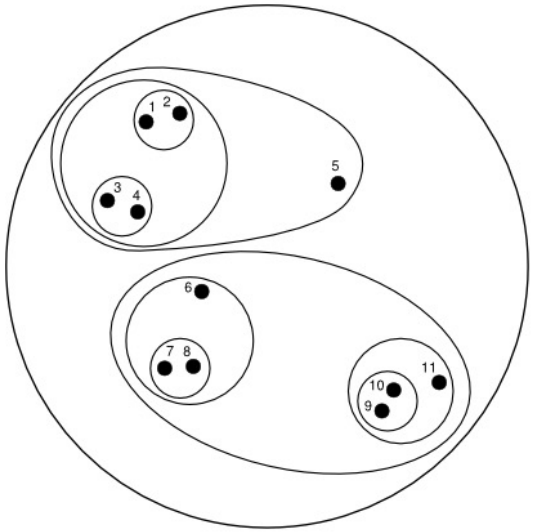


Hierarchical Clustering

Algorithm: Agglomerative (or Divisive)

- Start with one cluster per example
- Merge two nearest clusters
  - E.g., min, max, mean distance.
- Repeat until one cluster





Summary

- Clustering identifies typical groups.
- Need to understand the groups.
    - What do they have in common? Two options:
  - Manually examine elements of a cluster.
  - Use a supervised classifier!

1. Use the cluster labels as a supervision for a classifer.
2. Run the classifier, and examine the weights on each feature. The weights will say what is unique about each cluster.

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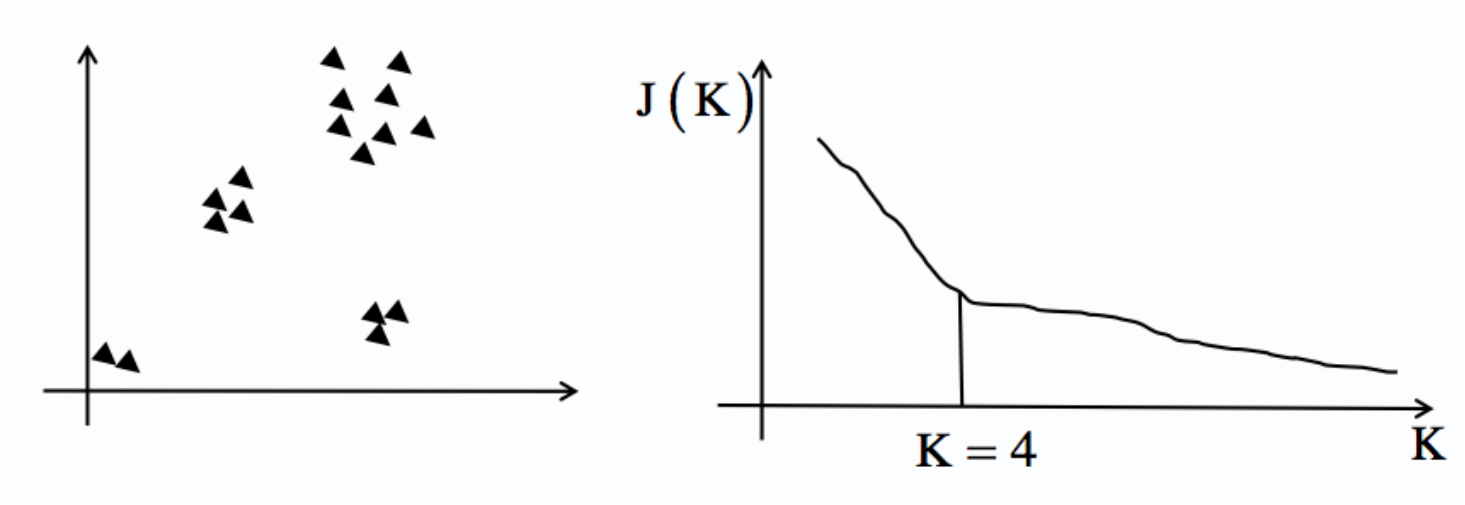
Choosing Number of Clusters

For K-means

- 1. Elbow Method
  - Plot $E_{KM/GMM}$ as a function of k, and choose the elbow point.



- 2. Present results to end users see what they prefer
  - Broader or more specialized groups

Choosing Number of Clusters

For K-means

3. Cross-validation
   - For K = 1…Large
     - Learn GMM/KM clusters on a train set.
     - Evaluate Quality(K) = quality on validation set.
   - Pick K with the highest validation set quality.

Choosing Number of Clusters

For K-means

- 4. BIC/AIC Criterion
  - (ML people: An approximation to the integration required in the Bayesian model selection)
  - Adds a penalty to the cost that penalises more complex models.
    - P: Number of parameters in model. N: Number of data points.
  - Evaluate modified cost E^K_{BIC} for many values of K.
  - Pick the K with best cost

$$E^K_{BIC} = E^K + \frac{p}{2} logN$$

$$E_{KM}(D, c_{1:N}, \mu_{1:K}) = \sum_{i=1}^{N} \left(x_i - \mu_{c_i}\right)^2$$

$$E_{GMM}(D; \boldsymbol{\pi}, \mathbf{u}, S) = -\log \prod_{i=1}^{N} \sum_{k=1}^{K} \pi_k N(\mathbf{x}_i; \mathbf{u}_k, S_k)$$

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  - Gaussian Mixtures
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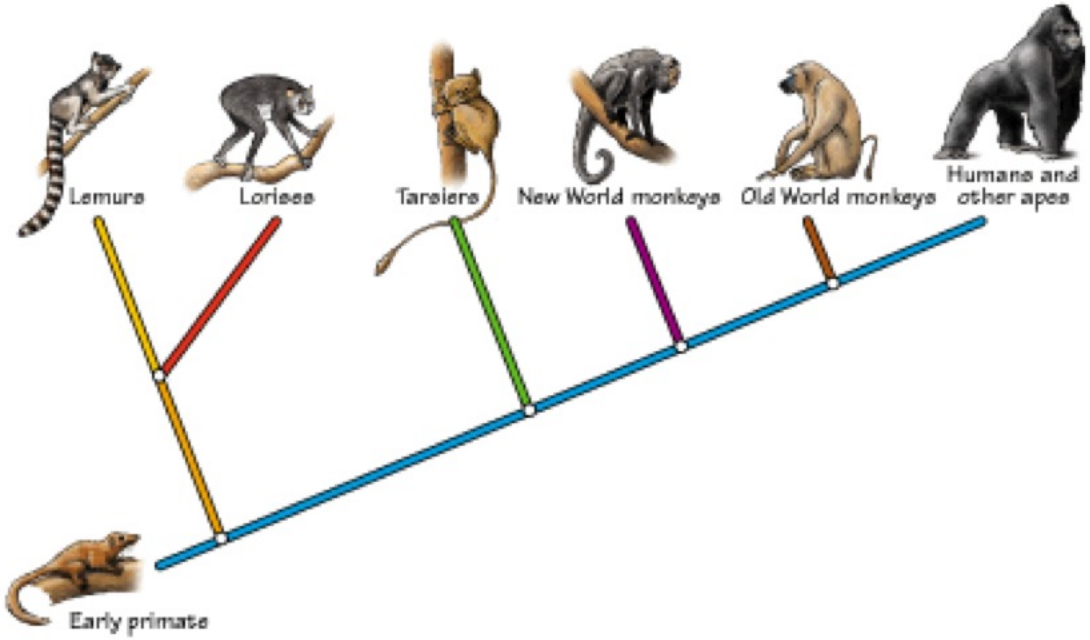
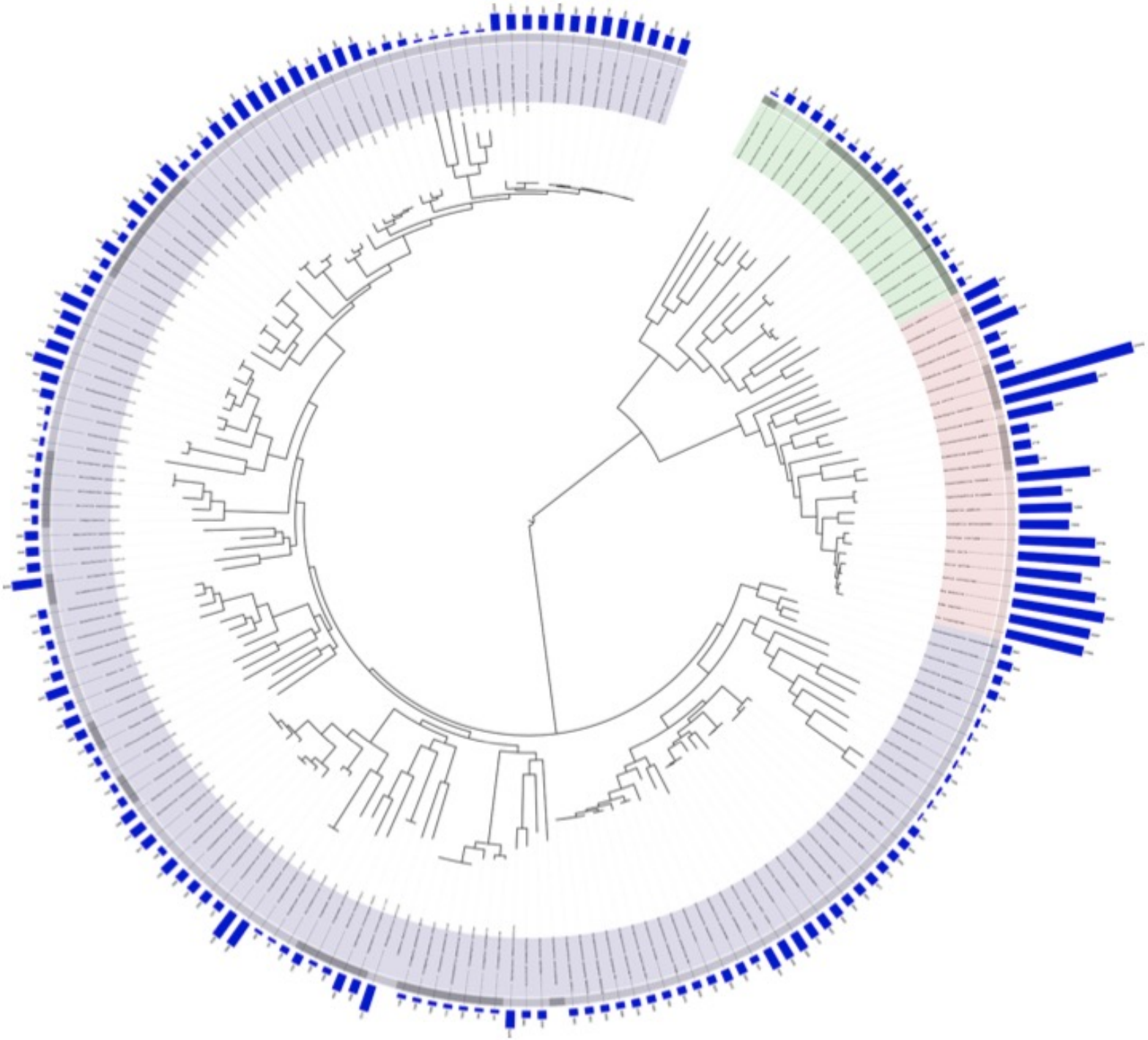
Applications: Bioinformatics

- Clustering of Gene Activity from Microarrays
  - Input: Gene activations
  - Output: Gene clusters
  - Discover which genes activate at the same time (appeared in the same cluster)
  - => Help discover relation between dissimilar genes

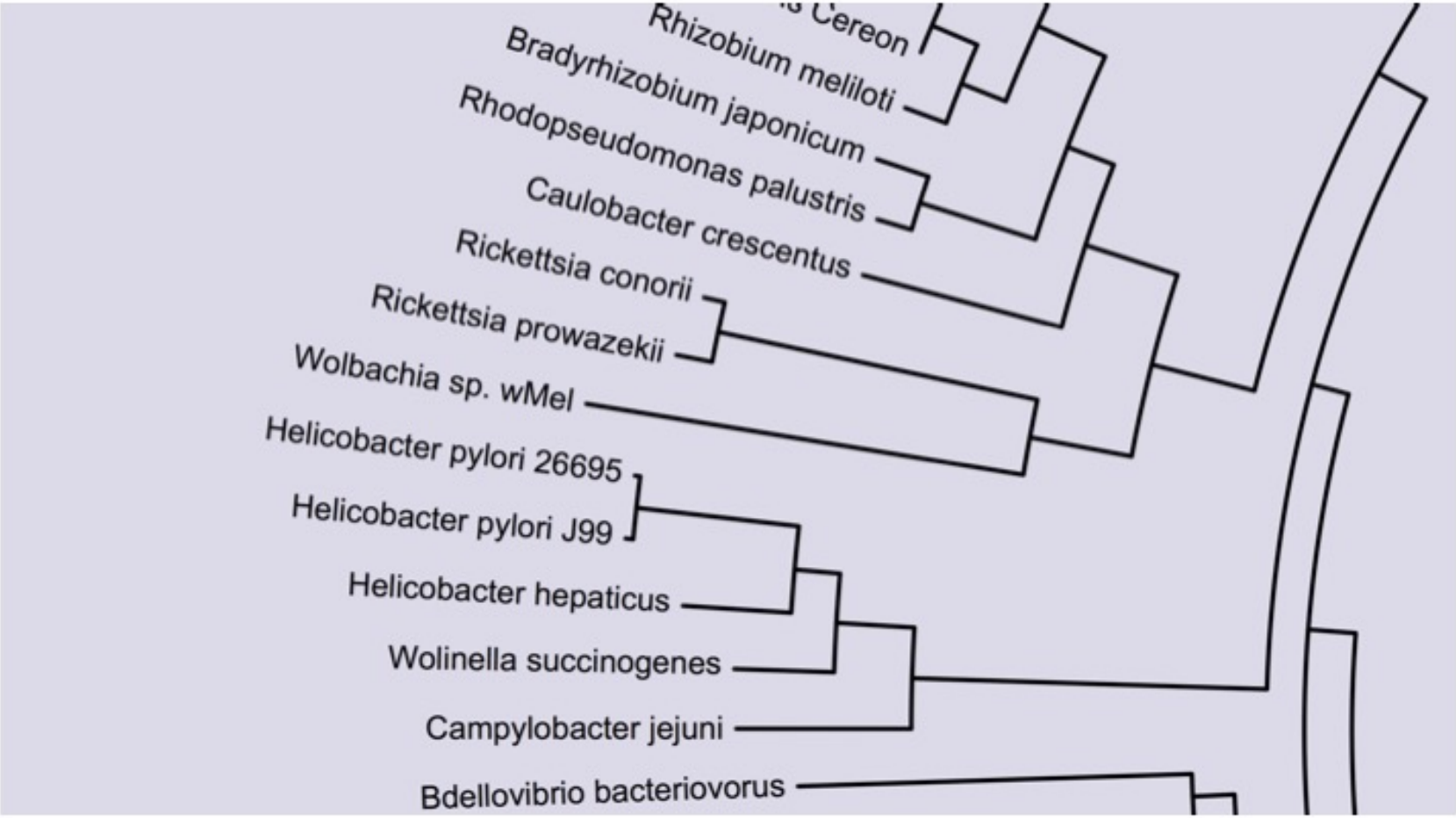
Applications: Bioinformatics for Evolutionary Biology

- Hierarchical Clustering of Genes:
  - Input: Genes of different species
  - Output: Dendrogram relating the genes
  - => Reveals evolutionary history





Evolutionary Biology



Applications: Finance

- Clustering companies
  - Input: Corporate data (e.g., financials)
  - Output: Clusters of similar companies
- Application:
  - Develop hedging/investment strategies
  - Predict if one stock price will rise or fall

Applications: News Summarisation

- Clustering News articles:
  - Input: One news article per row (E.g., as bag of words)
  - Output: Clusters of similar news articles.
- Application:
  - Get an overview of today's news by one article in each cluster.
    - No redundancy in stories

Application: Video Summarisation

EECS Research ☺

- Context:
  - Very many surveillance cameras recording long periods of video.
  - Exhaustively watching all is too time-consuming.
  - Want to get an overview of what happened during the hour/day/week
- Clustering
  - Input: Video frames/clips
  - Output: A category of every frame/clip

Application: Video Summarisation

EECS Research ☺

- Video Summarisation
  - A few clips quickly summarise the typical events
  - Everything else during the day is “more of the same”

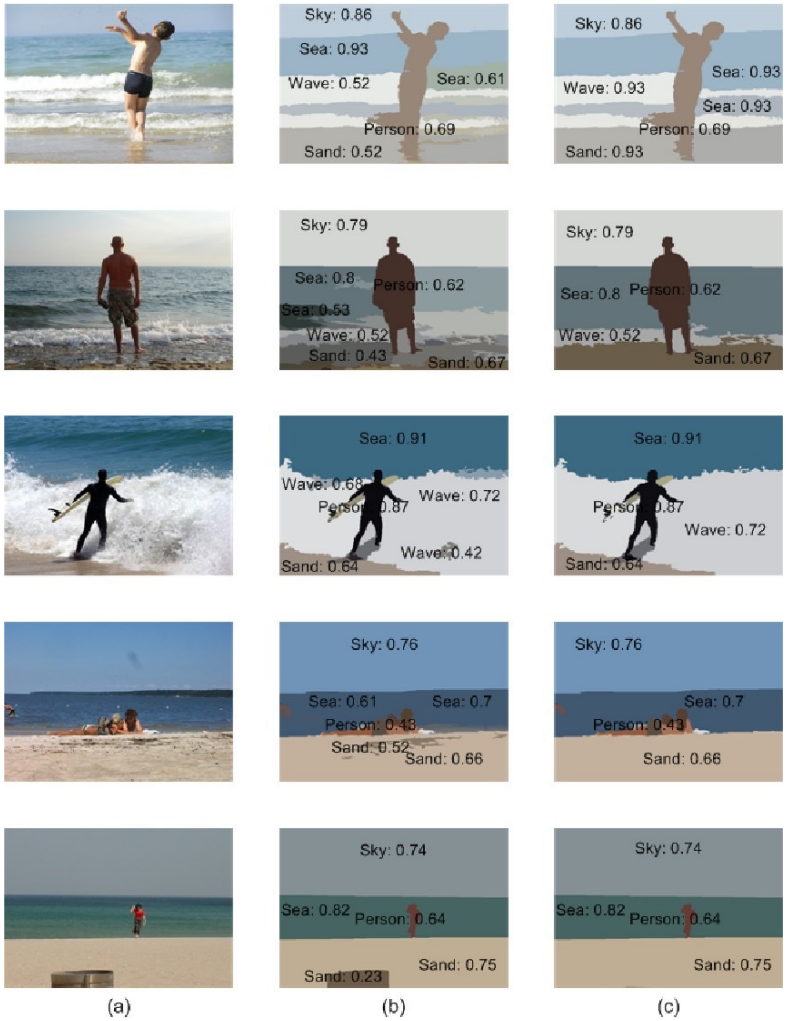
Applications: Recommendations (Content-based)

- Input:
  - Descriptions for each product
- Output:
  - Clusters of similar products
- Application:
  - Use discovered product similarity to choose a similar product to recommend

Applications: Image Processing

Photoshop filters…

- Input: Image Pixels
- Output: Segmentation (Grouping of all N pixels into K groups)



Applications: Politics

Clustering voting records

- Input: Votes of each MP
- Output: Clusters of voting patterns
  - Who tend to vote together / vote against each other.

You should know

- The idea and motivations for clustering
- Be able to sketch algorithms for:
  - K-means, hierarchical clustering
- Limitations of each algorithm
- Assess which of these algorithm would be suitable for a given problem